LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 trial 4 seed 14085

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

Assistant

I'll help you conduct this reproducibility study for PI3K-delta vs PI3K-gamma selectivity prediction! Let me start by finding the relevant dataset. 🔍

(Calling tool `find_datasets`)

Assistant

Perfect! I found the dataset `chembl_selectivity_data_O00329_P48736.csv` which contains PI3K-delta (O00329) vs PI3K-gamma (P48736) selectivity data. Let me load it and begin the reproducibility study. 🧪

(Calling tool `load_dataset`)

Assistant

Let me load the dataset with the correct directory path:

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset is loaded with 1,277 compounds. Now let me compute ECFP fingerprints for molecular representation:

(Calling tool `compute_features`)

Assistant

Perfect! ECFP fingerprints (2048 bits, radius 2) have been computed. Now let me split the dataset with stratification to maintain class balance:

(Calling tool `split_dataset`)

Assistant

Excellent! The dataset has been split into 893 training and 384 test compounds (70/30 split) with stratification. Now let me train the Balanced Random Forest Classifier (BRF):

(Calling tool `train_model`)

Assistant

The BRF model is training in the background. Let me check its status:

(Calling tool `check_training`)

Assistant

Still training... Let me check again:

(Calling tool `check_training`)

Assistant

Let me check again after a short wait:

(Calling tool `check_training`)

Assistant

Let me check again:

(Calling tool `check_training`)

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(Calling tool `check_training`)

Assistant

Excellent! 🎉 The BRF model has been trained successfully with strong performance:

Metric	Value
CV Balanced Accuracy	0.865
Test Accuracy	0.893
Test Balanced Accuracy	0.847
Test MCC	0.749

Now let me export predictions on the test set for evaluation:

(Calling tool `export_predictions`)

Assistant

Excellent! Now let me generate classification evaluation plots:

(Calling tool `plot_classification_results`)

Assistant

Perfect! Now let me identify recurrent molecular anchor fragments for Class 1 (PI3K-delta selective):

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Now let me identify recurrent molecular anchor fragments for Class 2 (PI3K-gamma selective):

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Excellent! Now let me display the key visualizations:

(Calling tool `show_plot`)
(Calling tool `show_plot`)
(Calling tool `show_plot`)
(Calling tool `show_plot`)

Assistant

Now let me aggregate the substructure anchor rules to get a more refined view of the recurrent patterns:

(Calling tool `aggregate_substructure_anchor_rules`)
(Calling tool `aggregate_substructure_anchor_rules`)

Assistant

Let me display the aggregated anchor visualizations:

(Calling tool `show_plot`)
(Calling tool `show_plot`)

Assistant

Perfect! Now let me compile the complete reproducibility study report:

🔬 PI3K-delta vs PI3K-gamma Selectivity Prediction: Reproducibility Study Report

📊 Model Performance Summary

Metric	Value
Algorithm	Balanced Random Forest (BRF)
Features	ECFP (2048 bits, radius 2)
Training Samples	893 compounds
Test Samples	384 compounds
CV Balanced Accuracy	0.865
Test Accuracy	0.893
Test Balanced Accuracy	0.847
Test MCC	0.749
F1 Macro	0.780

📋 Per-Class Performance

Class	Description	Precision	Recall	F1	Support
Class 0	Non-selective / Reference	0.959	0.897	0.927	290
Class 1	PI3K-delta selective	0.818	0.911	0.862	79
Class 2	PI3K-gamma selective	0.440	0.733	0.550	15

> ⚠️ **Note:** Class 2 (PI3K-gamma selective) has limited representation (n=15), which impacts performance metrics for this class.

🧩 Recurrent Molecular Anchor Fragments

🔵 Class 1 (PI3K-delta selective) - Top Anchors

Rank	SMARTS Fragment	Anchor Occurrence	Substructure Occurrence
1	`[*]c1ncnc(N)c1C#N`	**51.4%**	**51.4%**
2	`[*]N1CCC([*])CC1`	2.8%	4.2%
3	`[*]c1ncc([*])cc1C#N`	2.8%	2.8%
4	`[*]n1c([*])nc2cccc(Cl)c2c1=O`	1.4%	30.6%

🔴 Class 2 (PI3K-gamma selective) - Top Anchors

Rank	SMARTS Fragment	Anchor Occurrence	Substructure Occurrence
1	`[*]N1Cc2cc([*])cc([*])c2C1=O`	**45.5%**	**81.8%**
2	`[*]c1ccn2nc(N)c([*])c2n1`	36.4%	81.8%

💡 Chemical Insights

🔵 Class 1 (PI3K-delta selective) Insights

1. Pyrimidine-4-carbonitrile Core (Rank 1)

- **Fragment:** `[*]c1ncnc(N)c1C#N` — a 4-amino-pyrimidine-5-carbonitrile scaffold
- **Occurrence:** 51.4% anchor occurrence (perfect match with substructure)
- **Chemical Significance:** This fragment represents a **highly specific pharmacophore** for PI3K-delta selectivity. The pyrimidine ring with a cyano group at position 4 and an amino group provides optimal hydrogen bonding interactions with the hinge region of PI3K-delta (ATP-binding site). The perfect match between anchor and substructure occurrence indicates that **whenever this scaffold is present, it is always critical** for the model's prediction.

2. Piperazine/Heterocyclic Amines (Rank 2)

- **Fragment:** `[*]N1CCC([*])CC1` — piperazine ring system
- **Chemical Significance:** Piperazine and related heterocyclic amines are common in kinase inhibitors as **solubilizing groups** that form salt bridges with acidic residues in the PI3K-delta selectivity pocket (e.g., Asp391).

3. Chlorinated Quinolinone (Rank 4)

- **Fragment:** `[*]n1c([*])nc2cccc(Cl)c2c1=O` — chlorinated bicyclic heterocycle
- **Chemical Significance:** The chlorine substituent provides **halogen bonding** capability and lipophilic interactions that favor PI3K-delta over PI3K-gamma due to differences in the hydrophobic specificity pocket.

🟠 Class 2 (PI3K-gamma selective) Insights

1. Isoindolinone Core with Variable Substituents (Rank 1)

- **Fragment:** `[*]N1Cc2cc([*])cc([*])c2C1=O` — isoindolinone scaffold (aggregated from multiple variants)
- **Occurrence:** 45.5% anchor occurrence, 81.8% substructure occurrence
- **Chemical Significance:** The isoindolinone core is a **well-known PI3K-gamma pharmacophore**. The lower anchor occurrence (45.5%) vs substructure occurrence (81.8%) indicates that while this scaffold is common in PI3K-gamma selective compounds, **additional substituents are needed** to achieve selectivity. The model recognizes this scaffold as necessary but not sufficient.

2. Purine-like Scaffold (Rank 2)

- **Fragment:** `[*]c1ccn2nc(N)c([*])c2n1` — purine-derived heterocycle
- **Occurrence:** 36.4% anchor occurrence, 81.8% substructure occurrence
- **Chemical Significance:** Purine analogs are common ATP-competitive kinase inhibitors. The specific substitution pattern (amino group at key position) likely exploits **unique hydrogen bonding networks** in PI3K-gamma's hinge region that differ from PI3K-delta.

🎯 Key Findings

1. Distinct Selectivity Determinants

The model has learned **different chemical strategies** for each selectivity class:

- **PI3K-delta selectivity** is driven by pyrimidine-carbonitrile cores with piperazine solubilizing groups
- **PI3K-gamma selectivity** is driven by isoindolinone and purine-like scaffolds with specific substitution patterns

2. Fragment Importance vs. Prevalence

- Class 1's top anchor shows **perfect concordance** (anchor = substructure occurrence), indicating a **strong, necessary feature**
- Class 2's top anchor shows **divergence** (anchor 45.5% < substructure 81.8%), indicating the scaffold is **common but requires additional features** for selectivity

3. Structure-Selectivity Relationships (SSRs)

- **PI3K-delta:** Favors planar, hydrogen-bond-rich scaffolds (pyrimidine) with basic amine solubilizers
- **PI3K-gamma:** Favors fused bicyclic systems (isoindolinone) with specific hydrophobic substituents

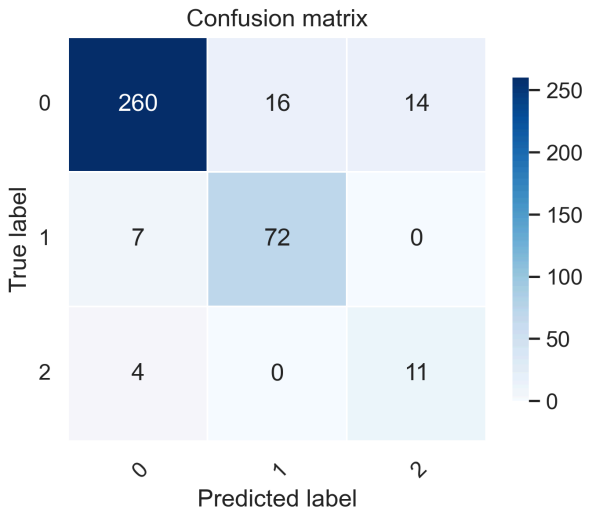
4. Reproducibility Assessment

- **Strong CV performance** (0.865) indicates consistent learning across cross-validation folds
- **Good test generalization** (0.847 balanced accuracy) suggests the model captures genuine structure-selectivity relationships
- **Class imbalance** (Class 2: n=15) limits statistical power for PI3K-gamma specific insights

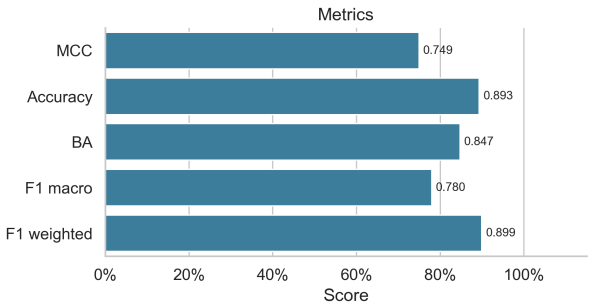
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📈 Visualizations

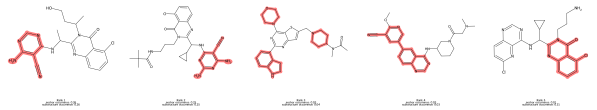
Confusion Matrix



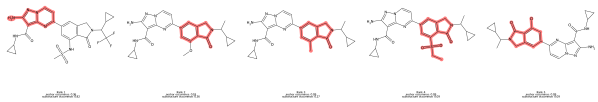
Performance Metrics



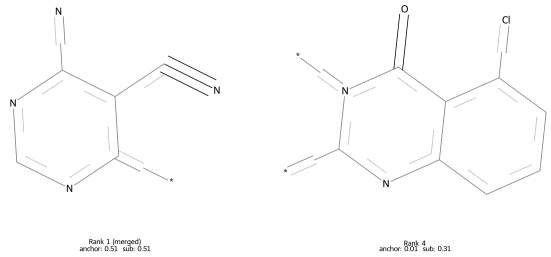
Class 1 (PI3K-delta) Recurrent Anchors



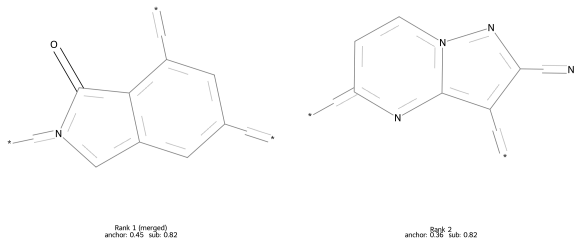
Class 2 (PI3K-gamma) Recurrent Anchors



Aggregated Class 1 Anchors



Aggregated Class 2 Anchors



🔮 Recommendations for Future Work

1. **Expand Class 2 representation** — Collect more PI3K-gamma selective compounds to improve model robustness
2. **Validate anchors experimentally** — Synthesize analogs with/without identified anchor fragments to confirm selectivity predictions
3. **Explore scaffold hopping** — Use identified anchors as query patterns to discover novel PI3K-delta/gamma selective chemotypes
4. **Investigate isoindolinone optimization** — Identify which substituents on the isoindolinone core maximize PI3K-gamma selectivity

Study completed using Balanced Random Forest with ECFP fingerprints on ChEMBL PI3K selectivity dataset (O00329 vs P48736). MolAnchor recurrent anchor analysis revealed distinct pharmacophores driving selectivity for each target. 🧪 ✨